

MediaOps CoPilot

Autonomous incident response for live video

When something breaks behind a live stream, it detects, diagnoses, fixes, and verifies recovery — autonomously, before viewers notice.

Gemini • Google Cloud • Grafana

The problem

A broken stream is detected in seconds. Fixing it, by hand, can take 15–30 minutes.

Before MediaOps CoPilot

01 Alert

An alert says something is wrong.

02 Investigate

An engineer manually checks video and telemetry.

03 Diagnose & fix

They find the cause, pick a fix, run it, and check recovery.

04 Wait

15–30 minutes pass while viewers see broken video.

With MediaOps CoPilot

01 Detect

An always-on monitor continuously checks the metrics and catches the problem automatically.

02 Diagnose

AI investigates the video and the telemetry in parallel.

03 Remediate

A tested fix is authorized and executed safely.

04 Verify

Recovery is independently confirmed — about 48 seconds.

For teams where every minute of downtime costs viewers, revenue, and engineering time.

How it's built

Every Google Cloud and Grafana integration runs live in the pipeline — invoked at runtime on each incident, not just cited.

Gemini via Google ADK

The Vision, Infra, and Remediation agents are built with the Agent Development Kit and call Gemini on Vertex AI at runtime; output is validated against a strict schema.

Vertex AI embeddings

Each incident is embedded and matched against past incidents with cosine similarity + a relevance threshold (RAG).

Firestore

Authoritative incident state, the idempotency ledger, and the growing knowledge base — persisted at every step.

Grafana + Prometheus

Agents read live telemetry read-only via Grafana MCP; metrics remote-write to a Grafana Cloud dashboard.

ffmpeg workload

A real encoding pipeline whose faults genuinely move the metrics, so a correct fix truly restores them.

Cloud Run

Single always-on container; credentials via the service account (ADC) — a real incident runs to CLOSED on the public URL.

Guardrails are plain deterministic code; AI is walled off from execution — the technical expression of "AI judgment, deterministic authority."

What it does

A full autonomous incident lifecycle — no human in the loop for routine failures.

1

Detect

Deterministic monitor watches telemetry continuously.

2

Diagnose

AI reads the video symptom and the infrastructure metrics.

3

Decide

Proposes a fix from a fixed set of tested actions.

4

Gate

Deterministic safety check authorizes or blocks it.

5

Execute

Runs the approved fix against the real infrastructure.

6

Verify

Independently confirms recovery in both domains.

Core principle: AI supplies judgment • deterministic code supplies authority

System architecture

Continuous monitoring → two AI agents investigate → confirm → safely fix (one retry if blocked) → verify

Always-on layer

A deterministic monitor watches the live metrics continuously and raises one incident per real problem.

AI investigation

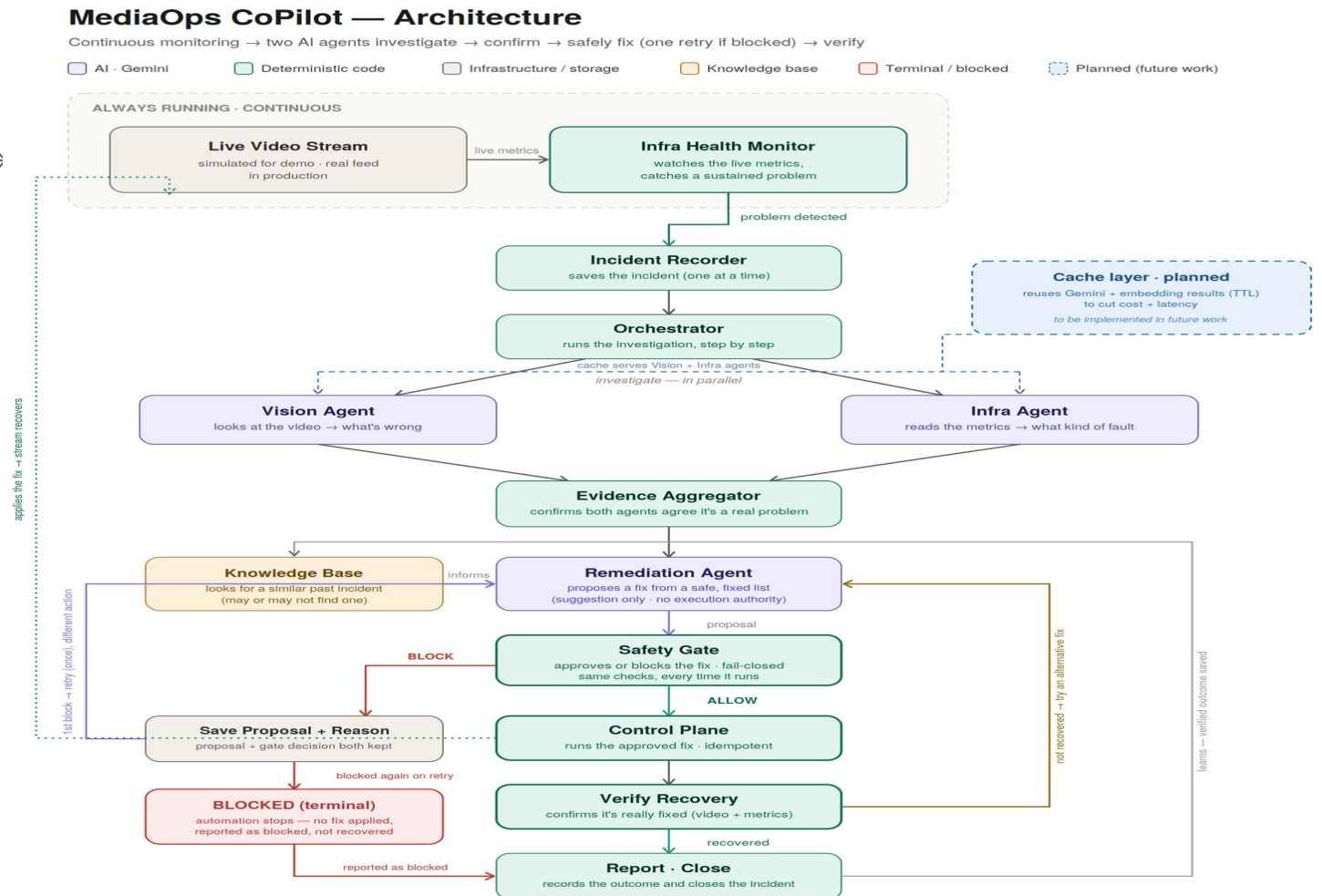
Two Gemini agents investigate in parallel — video and telemetry — and their findings must corroborate.

Guarded authority

A fail-closed Safety Gate authorizes the fix; the Control Plane executes it idempotently; recovery is verified.

Operational memory

Every verified recovery is saved back — the system learns from real incidents over time.



Safe by design

Autonomy is bounded by deterministic guardrails at every step.



Detection

Sustained-anomaly threshold — never fires on a transient blip.



Corroboration

Video and telemetry evidence must independently agree.



Fixed action set

AI can only choose from tested actions — it cannot invent commands.



Safety Gate

Deterministic, fail-closed — any error or bad state means BLOCK.



Idempotent execution

Duplicate delivery can never execute a fix twice.



Independent verification

"It ran" is not "it recovered" — recovery is proven, sustained.

The three AI agents

Each has one clear job — they investigate and recommend, but never act on their own.

1

Vision Agent

Looks at the actual video

Gemini reads a real frame from the stream and identifies the viewer-visible symptom — blocky, frozen, black, or fine. This is the "what does the viewer see?" evidence.

2

Infra Agent

Reads the infrastructure

Gemini interprets the live telemetry (CPU, frame rate, packet loss) via Grafana and classifies what kind of fault it is. This is the "what does the system say?" evidence.

3

Remediation Agent

Proposes the fix

Given both agents' evidence and similar past incidents, Gemini recommends one fix — chosen only from a fixed, tested set of actions. It proposes; it never executes.

Their findings are cross-checked: only when the Vision and Infra evidence agree is it confirmed a real incident.

Why monitoring isn't enough

Detection is easy. Understanding the cause and safely fixing it is the hard part — and that's what monitoring leaves to a human.

Monitoring stops here

Detects that something is wrong

An alert fires: "pipeline unhealthy."

...then it hands you a question, not an answer:

- What is the viewer actually seeing?
- What is the root cause?
- Which fix is safe to apply?

...and leaves the rest to humans

Diagnose

An engineer investigates video + telemetry to find the cause.

Decide

A human judges which of many fixes to apply.

Execute & verify

Someone runs the fix and confirms it actually worked.

Detection is automated. Recovery is not. MediaOps CoPilot closes the gap — **detect → diagnose → safely fix → verify, autonomously.**

Tested and guarded at every step

The AI is bounded by deterministic guardrails — and every component is tested against known faults.

Detection

Only a sustained problem raises an incident — never a one-off blip.

Evidence must agree

Video and telemetry are cross-checked; contradictory evidence is rejected.

Fixed action set

The AI can only choose tested actions — it cannot invent a command.

Safety Gate

A deterministic, fail-closed check approves or blocks every fix. Any error means block.

Safe execution

The approved fix runs once and only once — no duplicate actions.

Independent verification

Recovery must be proven in both video and metrics — "it ran" is not "it's fixed."

Every component is validated against a golden set of known faults, run repeatedly — and the AI agents reach 98% accuracy, with uncertain results caught by the guardrails rather than acted on.

The impact

~48s

average time to recovery, fully autonomous

15–30 min

typical manual incident response

- Fewer viewer-impacting outages
- Engineers freed from repetitive incident detective work
- Consistent, auditable handling that improves over time

**When something goes wrong behind the stream,
the viewer shouldn't have to know.**

MediaOps CoPilot — an autonomous first responder for media infrastructure.

Live • Gemini • Google Cloud • Grafana